

LIFE SCIENCES

LIFE SCIENCES GRADE 12 SOURCE-EMBEDDED SMOKE TEST

SUBJECT	LIFE SCIENCES
GRADE	12
ASSESSMENT TASK	LIFE SCIENCES GRADE 12 SOURCE-EMBEDDED SMOKE TEST
MARKS	27
TIME	45 minutes

Educator approval required before classroom use.

INSTRUCTIONS AND INFORMATION

1. Read all instructions carefully.
2. Answer all questions or complete all activities as instructed.
3. Show all working where calculations, planning, diagrams or explanations are required.
4. Use the space provided by the educator, answer booklet or learner task book.
5. Write neatly and clearly.

SOURCE MATERIAL

SOURCE 1: Official-paper exemplar: Data Table

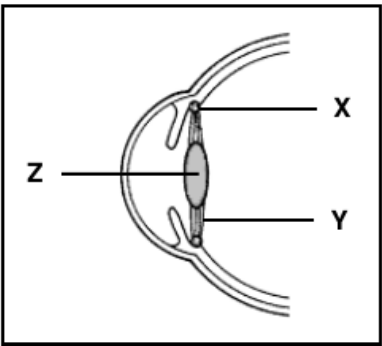
- 2.2 The table below shows the levels of two hormones during the menstrual cycle of a healthy human female.

DAY OF MENSTRUAL CYCLE	OESTROGEN LEVEL (pg/ml)	PROGESTERONE LEVEL (ng/ml)
4	55	0,23
8	70	0,03
10	280	0,03
12	300	0,03
14	140	3,0
16	110	12,5
20	80	15,0
24	70	5,0
28	65	0,8

- 2.2.1 On which day of this menstrual cycle is the level of progesterone the highest? (1)
- 2.2.2 Name the reproductive hormone that will begin to increase from day 24 of this female's menstrual cycle. (1)
- 2.2.3 Use data from the table to explain your answer to QUESTION 2.2.2. (2)
- 2.2.4 Calculate the percentage increase in the oestrogen level from day 8 to day 10. Show ALL your working. (3)
- 2.2.5 How would the progesterone level differ after day 20 if this female was pregnant? (1)

SOURCE 2: Official-paper exemplar: Diagram Or Model

1.1.6 The diagram below represents a part of the human eye.



A spectator in a soccer stadium is seated 200 metres away from the field.

Which ONE of the following describes the condition of structures X, Y and Z when he looks at the ball placed in the middle of the field?

	MUSCLE IN X	STRUCTURE Y	PART Z
A	Relaxed	Slack	More convex
B	Contracted	Slack	Less convex
C	Relaxed	Taut	Less convex
D	Contracted	Taut	More convex

1.1.7 Which ONE of the following is involved in thermoregulation?

- A Corpus callosum
- B Hypothalamus
- C Cerebellum
- D Spinal cord

Paper: Life Sciences P1 November 2024 | Page: 4 | Source Type: diagram_or_model

SOURCE 3: Official-paper exemplar: Graph Or Chart

described in QUESTION 2.3.2.

11

2.3.4 Draw a bar graph to show the results in the table.

(6)

(14)

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VISUAL MATERIAL

FIGURE 1: Investigation Data And Method Sheet

Life Sciences		INVESTIGATION	
Life Sciences Grade 12 Source-Embedded Smoke Test			
Investigation method, data and conclusion			
Aim Variables Method Results Conclusion			
Reading	Observation	Unit	Pattern / anomaly
Conclusion supported by data:			

QUESTIONS

QUESTION 1: SOURCE INTERPRETATION

Instructions: Use the supplied source material. Answer from visible evidence and relevant subject knowledge.

1.1. Refer to SOURCE 1. Identify one important feature of the data table and explain how it could be used as evidence in Life Sciences. **(4)**

1.2. Refer to SOURCE 2. Identify one important feature of the diagram or model and explain how it could be used as evidence in Life Sciences. **(4)**

1.3. Refer to SOURCE 3. Identify one important feature of the graph or chart and explain how it could be used as evidence in Life Sciences. **(4)**

1.4. Compare TWO of the supplied sources. Explain which one would produce the strongest assessment question and why. **(5)**

QUESTION 2: METHOD AND REVIEW

Instructions: Evaluate whether this generated assessment pack is ready for a real learner group.

2.1. State TWO checks an educator must perform before releasing this assessment to learners. **(4)**

2.2. Explain why official-paper source exemplars are stronger than generic generated visuals for this route. **(6)**

MARKING GUIDELINE

TASK 1: SOURCE INTERPRETATION

1.1. Refer to SOURCE 1. Identify one important feature of the data table and explain how it could be used as evidence in Life Sciences. **(4)**

- Relevant feature from SOURCE 1 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.2. Refer to SOURCE 2. Identify one important feature of the diagram or model and explain how it could be used as evidence in Life Sciences. **(4)**

- Relevant feature from SOURCE 2 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.3. Refer to SOURCE 3. Identify one important feature of the graph or chart and explain how it could be used as evidence in Life Sciences. **(4)**

- Relevant feature from SOURCE 3 is identified.
- Explanation is linked to the subject context.
- Answer uses visible evidence rather than unsupported recall.

1.4. Compare TWO of the supplied sources. Explain which one would produce the strongest assessment question and why. **(5)**

- Two supplied sources are compared.
- A justified judgement is made.
- Reasoning mentions readability, data richness, visual clarity or subject fit.

TASK 2: METHOD AND REVIEW

2.1. State TWO checks an educator must perform before releasing this assessment to learners. **(4)**

- Check source readability and relevance.
- Check CAPS topic/grade alignment.
- Check mark allocation and memo correctness.
- Check copyright/reuse conditions for source material.

2.2. Explain why official-paper source exemplars are stronger than generic generated visuals for this route. **(6)**

- They reveal actual source types used in exams.
- They reduce invented visual nonsense.
- They still require adaptation and educator approval.

RUBRIC

Criterion	Marks	Descriptor
Source evidence use	10	Answers use visible evidence from supplied sources.
Subject accuracy	10	Answers use correct subject concepts and terminology.
Review readiness	10	Risks and approval checks are identified clearly.

EDUCATOR REVIEW CHECKLIST

- Confirm each embedded source is readable after PDF export.
- Confirm source use/copyright status before client delivery.
- Confirm questions refer only to supplied or clearly stated evidence.
- Confirm memo and mark allocation match the subject route.